

Worksheet 3: Decimals – 2nd ESO

1. Theory

Decimals are used to represent fractions and non-integer values precisely. In FM radio, each station has a specific frequency, measured in MHz, with decimals indicating the exact position on the spectrum.

Rounding

Sometimes decimals are rounded for easier calculations or due to technical limitations, like when the frequency is displayed on the screen.

Ordering

Knowing how to order decimals is important for station lists or data processing.

2. Exercises

Rounding

Round the following frequencies to one decimal place:

- 97.645 MHz
- 102.387 MHz
- 88.993 MHz

Order the following frequencies from smallest to largest:

- 101.3 MHz
- 98.7 MHz
- 100.05 MHz
- 99.99 MHz

Convert to Hz:

- 96.7 MHz
- 103.25 MHz

3. Competence-based problems

Problem 1

If a digital radio only displays one decimal digit, how does this affect the frequency precision? Calculate the maximum possible difference between the real and displayed frequency if rounded to one decimal place.

Problem 2

A student tunes the radio to frequency 99.85 MHz. How many Hz does this frequency equal?

Solutions

Rounding:

- $97.645 \rightarrow 97.6 \text{ MHz}$
- $102.387 \rightarrow 102.4 \text{ MHz}$
- $88.993 \rightarrow 89.0 \text{ MHz}$

Ordering:

$98.7 < 99.99 < 100.05 < 101.3 \text{ MHz}$

Conversion:

- $96.7 \text{ MHz} = 96,700,000 \text{ Hz}$
- $103.25 \text{ MHz} = 103,250,000 \text{ Hz}$

Problems:

- 1. With one decimal, the frequency can vary by $\pm 0.05 \text{ MHz}$ (50 kHz), which may affect reception.
 - 2. $99.85 \text{ MHz} = 99,850,000 \text{ Hz}$
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